Beyond the hourly average

Quarter-hour price reconstruction for AET

The challenge

Our assigned AET challenge was to reconstruct **96 quarter-hour electricity prices from 24 supplied hourly prices**, for each of four market zones. The four reconstructed prices within every hour must average back to the supplied hourly price, within a tolerance of **0.0001 €/MWh**.

An hourly average hides movements between its four quarters. This matters because electricity is not necessarily delivered evenly throughout the hour, so a more detailed price curve could support more accurate valuation. Our task was to recover as much of that missing shape as possible using the official price data, while keeping the hourly reference unchanged.

Approach

Our analysis of the official data shaped the model. The hourly price already explains 93–96% of the price variance; what is scored is the shape inside each hour. That shape is largely predictable: the slope between neighbouring hours correlates 0.76–0.91 with the tilt inside the hour, as prices slide with the daily curve and snap back at the next hour. What remains concentrates at ramps, turning points and day edges. We therefore built the prediction in layers:

$$\hat{p}_{h,q} = \underbrace{b_{h,q}}_{\text{PCHIP baseline}} + 0.8 \underbrace{\left(c_h^{\text{slope}}\, s_q + c_h^{\text{curv}}\, k_q + c_h^{\text{zig}}\, z_q\right)}_{\text{LightGBM correction}}, \quad \text{then three fixed rules.}$$

Baseline. We interpolate the running sum of hourly prices with PCHIP (Piecewise Cubic Hermite Interpolating Polynomial) and difference it back into quarters. The increments telescope, so every hourly average is exact, and PCHIP does not overshoot at spikes. This captures the smooth part of the slope (development MAE 8.13 flat → 5.62 €/MWh).

Correction. With the hourly mean fixed, four quarters have exactly three degrees of freedom. Three orthonormal vectors whose entries sum to zero therefore describe every possible shape, and adding them can never move the average:

- **Slope**, $s = (-3, -1, 1, 3)/\sqrt{20}$: whether prices rise or fall within the hour.
- **Curvature**, $k = (1, -1, -1, 1)/2$: whether the middle quarters sit above or below the outer quarters.
- **Zigzag**, $z = (-1, 3, -3, 1)/\sqrt{20}$: alternating movements between quarters.

One LightGBM model per vector predicts its coefficient from the day's hourly curve, differences between zones, calendar information and earlier quarter-hour patterns, never from the target day's quarter-hours. Boosted trees suit the non-linear situations behind the remaining error, such as ramps and turning points, and an absolute-error loss keeps spikes from dominating training. This basis beat a Haar basis (3.98 vs 4.00 €/MWh), and removing cross-zone features worsened MAE to 4.04. Applying 80% of the predicted correction lowered MAE by a further 0.9% (6 of 8 folds): when the model is unsure, a partial correction is safer than a full one.

Fixed rules. Three rules with no fitted parameters then adjust each hour, keeping its average:

- **Zone coupling:** zones with almost the same hourly price (within 0.1 €/MWh) usually move together, so they share one shape: the average of their predicted shapes.
- **Partial tie:** zones within 5 €/MWh behave similarly but less reliably; each moves 40% towards the group's shape.
- **Sign rule:** quarters rarely change sign within an hour. In a positive hour no quarter may fall below −0.01 €/MWh, in a negative hour none may exceed 0; the remaining quarters shift to keep the average.

Adding the rules and 1000 (not 500) boosting rounds cut development MAE by 1.5% (all 8 folds). A final adjustment removes floating-point drift, and automated checks confirm the 0.0001 €/MWh tolerance.

Challenges and solutions

Working with incomplete information. Many quarter-hour profiles share the same hourly average, so we could learn recurring patterns but never recover every movement exactly.

Keeping the comparison fair. We used chronological validation and historical inputs matched by calendar date, so missing dates cannot shift the history and the target day's quarter-hours never reach the model. A change was promoted only if it lowered development MAE by at least 0.5%, improved at least two-thirds of the folds and left no zone more than 2% worse.

Deciding when to stop adding complexity. We tested Ridge, CatBoost, temporal CNN, dilated TCN, LSTM and Transformer models; for the correction we compared LightGBM with CatBoost, Ridge, Random Forest and Extra Trees, and we tried other interpolation methods, correction weights and per-zone models. Most ideas failed the gate. Local-geometry features slightly worsened our previous model's development MAE (**3.9479 to 3.9497 €/MWh**). We also explored temporal and Bayesian graph models, error-correction models and ensembles: in a separate **June–August window**, a 25% compact-graph / 75% previous-model blend reduced MAE from **4.2246 to 4.2058 €/MWh** (0.4%), below the gate and not worth the added complexity.

Results

We report mean absolute error (MAE) between reconstructed and observed prices, with equal weight for each day and zone. For development, we used eight chronological January–August folds, training on earlier dates and scoring days 19 through month-end. We then assessed a model trained through 31 August on 19–30 September. September results are also available for the graph experiments, and we scored September again before switching to the final rules, so we report this as a historical assessment rather than an untouched final test.

Overall reconstruction accuracy

Method	Development MAE	September MAE
Flat hourly prices	8.13	10.90
PCHIP	5.62	6.93
Full correction (scale 1.0)	3.98	5.43
Previous model (no rules)	3.95	5.04
Final model	**3.89**	**5.00**

Table 1: Average price error in €/MWh. Each step adds one layer of the approach; the full correction and the previous model (500 boosting rounds, no rules) are comparisons, not the submitted model.

By zone (September assessment)

Zone	Model MAE	Flat MAE	Error reduction
ZONE_0	3.83	10.78	64.5%
ZONE_1	6.12	13.30	54.0%
ZONE_2	5.49	9.72	43.5%
ZONE_3	4.57	9.79	53.3%

Table 2: September average price error by zone, comparing the final model with repeating the hourly price.

The final model reduces average price error by **52.2% on development dates** and **54.1% on the September assessment**, compared with flat prices. On development dates, PCHIP removes 31% of the flat error, and the correction and rules remove a further 31% of what remains. Gains follow the EDA: ZONE_0, where the slope between hours predicts the tilt best ($r = 0.91$), improves most; ZONE_2, where it is weakest ($r = 0.76$), improves least. Compared with the previous model, the September gain is smaller (0.8%), and ZONE_3 is slightly worse (4.57 instead of 4.55).

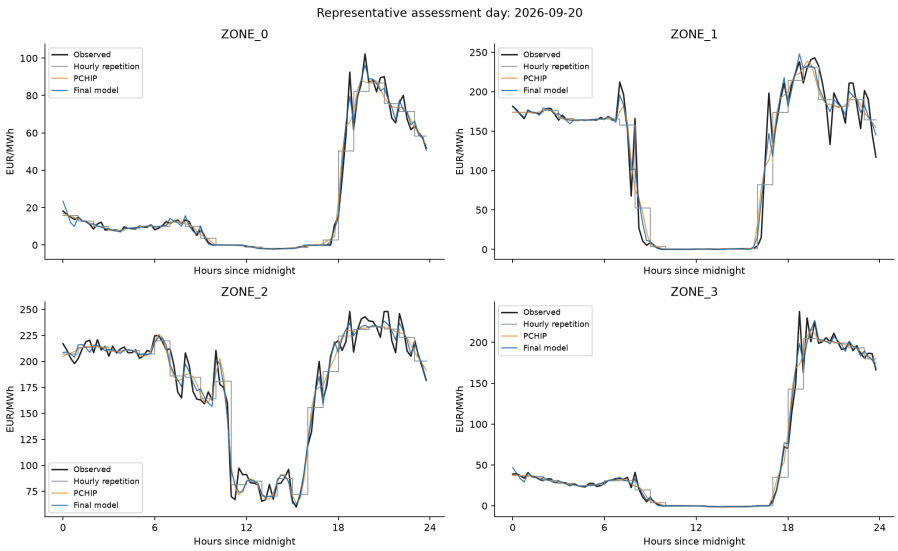


Figure 1: Reconstruction on 20 September 2026 across the four zones. Observed quarter-hour prices are compared with hourly repetition, PCHIP and the final model.

For us, these results suggest that there is useful structure to recover beyond the hourly average. However, we evaluated price accuracy; we have not measured the effect on actual trading results or portfolio valuations.

Dashboard: results and how to read them

We also wanted to make the results approachable for someone who does not work with machine learning. The main page of our interactive dashboard, **Model Results**, replays the September hold-out (or the user's own test days): for one zone or all four, and any date range, it compares reconstructed with observed prices and reports the error per zone. Three complementary pages help interpret these results:

- **Understanding the Model:** what the model relies on overall, and one hour broken down into baseline, slope, curvature, zigzag and rules, next to the observed prices (Figure 2).
- **Risk Alerts:** the hours to trust less, flagged before their 15-minute prices are known.
- **Data Story:** the patterns in the official data that shaped the model.

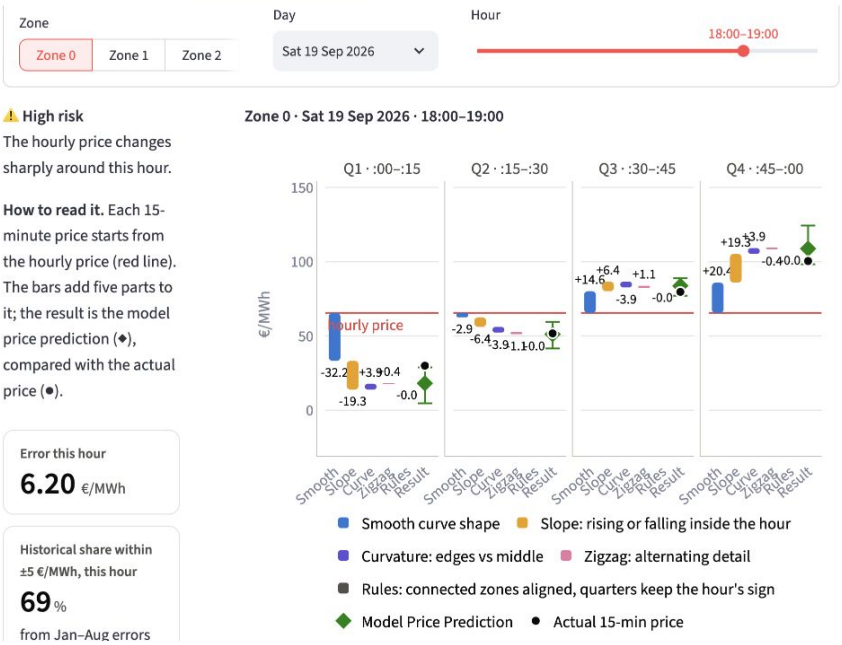


Figure 2: Understanding the Model page for ZONE_0 on 19 September 2026, 18:00–19:00.

The risk flags use hourly ramps, daily price variability and recent historical reconstruction difficulty, and name the main signal behind each label. They are **heuristics, not calibrated confidence probabilities**.

Does the indicator identify harder hours?

On the January–August development folds, hours flagged as high-risk had **2.8–3.9 times the reconstruction error** of low-risk hours across the four zones. This suggests that the indicator helps distinguish harder hours, although it does not guarantee an individual prediction's accuracy.

Zone	Low-risk MAE	High-risk MAE	High / low
ZONE_0	1.51	5.89	3.9×
ZONE_1	2.80	8.32	3.0×
ZONE_2	2.03	5.68	2.8×
ZONE_3	2.22	7.20	3.2×

Table 3: Observed reconstruction MAE in €/MWh by risk group on the development folds.

AI agents and LLM usage

We worked with LLM-based coding assistants throughout the project, especially when coding. It was an iterative process: we tried an approach, looked at the results, and used the assistants to help identify errors and work through possible solutions. We also discussed our EDA findings with them to explore which features or model changes might be worth testing. We then checked those suggestions through tests and backtesting, keeping the changes that improved the results. Finally, we used them to help organise and revise the repository, report and presentation, refining the explanations through several rounds of feedback.

More Information

The papers, libraries and challenge materials we consulted are listed in `docs/REFERENCES.md` in the repository, including references related to the approaches we tested but did not retain.